

PROJECT DOCUMENTATION

MERN STACK BOOK STORE

System Design, Implementation & Verification Walkthrough

DEVELOPMENT TEAM & ACADEMIC ROLES:

1. Roopesh Araja - Technical Architecture, Entity Relation Diagram, Features, Roles
2. Palla Satya Ramanaidu - MVC Pattern, Creating Project Folder, Client Setup
3. Bandi Sravya - Server Setup, Backend Structure, Development, Configure MongoDB
4. Vijaya Raju Jakkamsetti - Database Connection, Schema & Models, Frontend Structure
5. Vankayala Jaya Chandra Srinivas - Steps for Execution, Demo Screenshots, Drive Link

2. Project Overview

Purpose & Goals

The BOOK STORE (BookVerse) application is a full-stack e-commerce web platform built utilizing the MERN stack (MongoDB, Express.js, React.js, and Node.js). It provides a responsive interface where users can search, browse, wishlist, and buy books. It handles role-based logins separating standard customers, sellers/vendors, and system administrators.

Key Product Features

- **Role-Based Registration & Login:** Tabbed credentials security separating customers, sellers, and system administrators using JWT.
- **Seller Catalog Controls:** Approved sellers can list new books, upload cover art files via Multer, modify details, and check order statuses.
- **Customer Shop View:** Customers browse book cards, search by text, filter by genre, add items to a local wishlist, and check out.
- **Fulfillment Tracker:** Sellers monitor orders containing their items and toggle shipping status (ontheway, delivered).
- **Admin Dashboard:** Administrators view system aggregates, inspect an SVG bar chart of database counts, approve sellers, edit user details, and cancel orders.

3. Technical Architecture

Frontend (React Client)

The frontend UI is developed with React.js using Vite. Global authentication contexts manage user sessions and tokens, and Bootstrap handles responsive viewport scaling.

Backend (Node.js & Express.js)

The backend API uses Node.js and Express.js styled under the Model-View-Controller (MVC) design pattern. Mongoose manages DB transactions, Multer handles image uploads, and Bcryptjs handles password encryption.

Database Schema (MongoDB)

- **Admins:** Stores administrator name, credentials, and self ID.
 - **Sellers:** Stores vendor name, credentials, and isApproved boolean state.
 - **Users:** Stores customer name, credentials, and timestamps.
 - **Books:** Stores title, author, genre, price, stock quantity, cover image path, and seller references.
 - **Orders:** Stores customer name, delivery address, booking date, delivery target, price, and status (ontheway, delivered).
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4. Setup & Setup Instructions

Prerequisites

- Node.js (v16+) and npm installer installed.
- A MongoDB Atlas cluster configured with network whitelisting enabled.

Installation

- 1. Extract/clone the repository folder structure.
- 2. In backend/ folder, create a .env file containing:

```
PORT=8000
MONGO_URI=mongodb+srv://23pa1a5761_db_user:AR15l56xvSCDDLmM@book-store.ouo6cb6.mongodb.net/bookverse?appName=Book-Store
JWT_SECRET=bookversesecrettokenkey12345
```

- 3. Run "npm install" inside both backend/ and frontend/ folders to download packages.

5. Folder Structure

- **frontend/src/App.jsx**: Coordinates browser routes.
 - **frontend/src/context/AuthContext.jsx**: Manages active roles and API connections.
 - **frontend/src/components/**: Layout components (Navbar.jsx, OrderModal.jsx).
 - **frontend/src/pages/**: Dashboard and catalogue pages.
 - **backend/server.js**: Core Node Express entry point.
 - **backend/seed.js**: Clear and populate MongoDB collections.
 - **backend/controllers/**: Express business logic controllers.
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6. Running the Application

Start the local development servers using two separate terminals:

Backend Service (Port 8000)

```
cd backend  
npm start
```

Frontend React App (Port 5173)

```
cd frontend  
npm run dev
```

7. API Documentation

- **POST /api/users/register:** Create customer account.
- **POST /api/users/login:** Verify customer credentials.
- **GET /api/users/books:** List books. Supports query filters (?search=text&genre=fiction).
- **POST /api/users/orders:** Checkout. Body: { bookId, customerName, address, price }.
- **POST /api/seller/books:** Add new product listing (Multipart file attachment).
- **PUT /api/admin/sellers/approve/:id:** Toggle vendor isApproved to true.

8. Authentication & Security

- **Hashing:** Passwords are encrypted with Bcryptjs using 10 salt rounds.
- **JWT Bearer:** Authorizations are checked by parsing the Authorization Bearer header.

9. User Interface

- **Home Page:** Styled with a cream background, featured category selectors, and contact information.
 - **Admin Dashboards:** View database statistics (Users, Vendors, Items, Orders) and SVG bar charts.
 - **Seller Panels:** Books list, add-book forms, and order tracker panels.
 - **Customer Workflows:** Catalog grid, split Description/Info detail views, and My Orders list.
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10. Testing Strategy

- **Database Seed Check:** Seeder validates direct database clear and population.
- **Model Query Integrity:** `verify_backend.js` script queries Mongoose schema directly to verify record counts on Atlas.
- **Production Bundler Compile:** "npm run build" compiled with zero warnings.

12. Known Issues

- Atlas Network whitelist requires 0.0.0.0/0 to allow global connections.
- Multer image uploads are limited to 5MB.

13. Future Enhancements

- Integrate Stripe checkout API for credit card transaction processing.
- Incorporate email notifications via Nodemailer for low stock alerts.

11. Screenshots

Live Project Demo Link: <https://book-store-nzxy.onrender.com/>

1. Guest Home Page

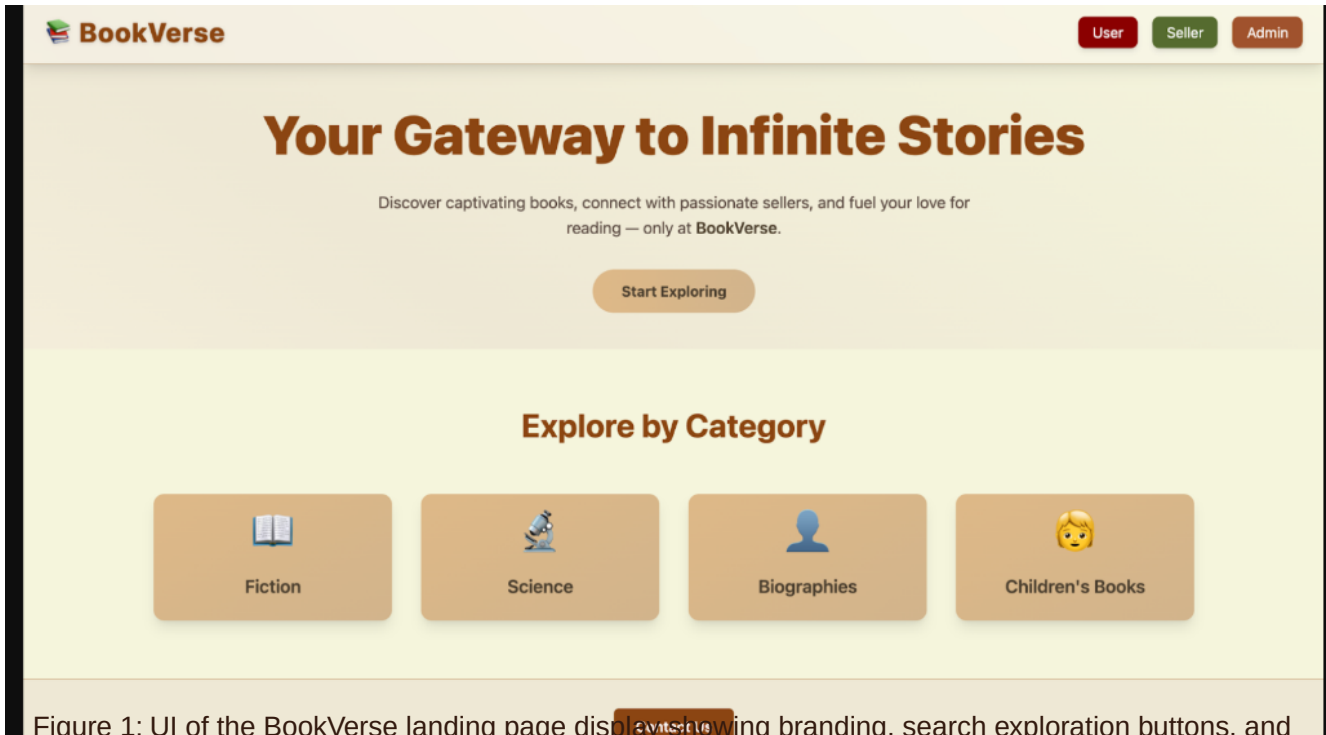
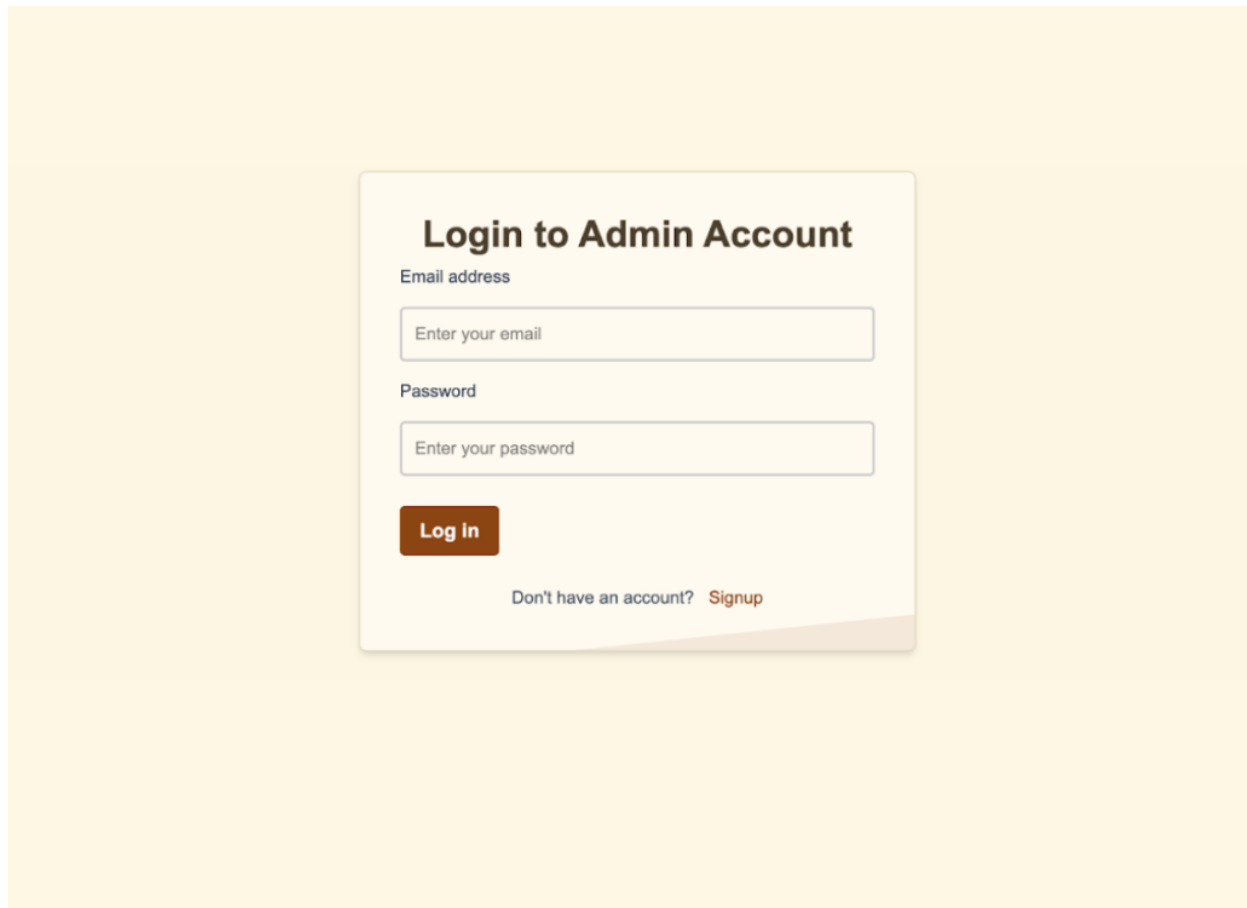


Figure 1: UI of the BookVerse landing page displays branding, search exploration buttons, and guest portal entry badges.

2. Admin Login Page



The image shows a login form titled "Login to Admin Account" centered on a light yellow background. The form is a white card with a subtle shadow. It contains two input fields: "Email address" with a placeholder "Enter your email" and "Password" with a placeholder "Enter your password". Below the password field is a dark brown "Log in" button. At the bottom of the card, there is a link that says "Don't have an account? Signup", where "Signup" is in a dark brown color.

Login to Admin Account

Email address

Enter your email

Password

Enter your password

Log in

Don't have an account? [Signup](#)

Figure 2: Authentication card panel interface for the system administrator login portal.

3. Seller Dashboard

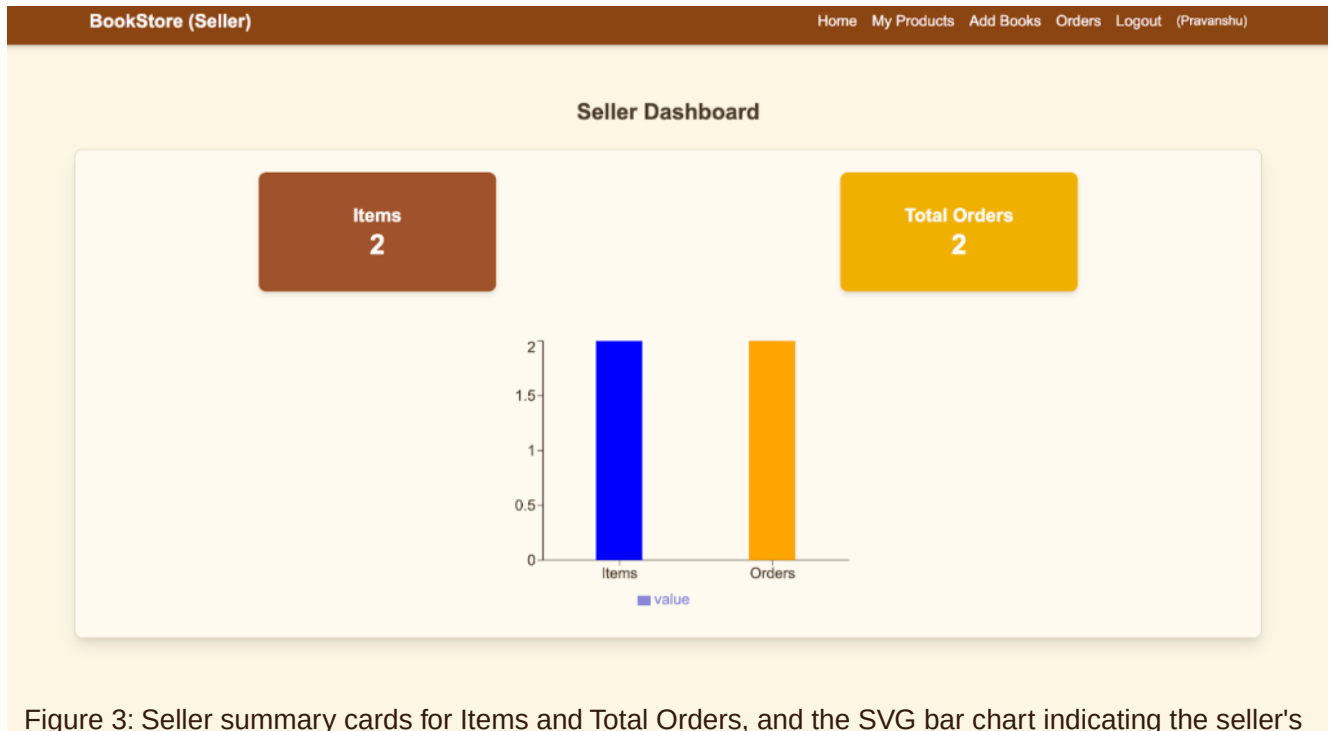


Figure 3: Seller summary cards for Items and Total Orders, and the SVG bar chart indicating the seller's metrics.

4. Seller Products List

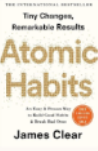
BookStore (Seller)

HomeMy ProductsAdd BooksOrdersLogout (Pravanshu)

Books List



The Alchemist
Author: Paulo Coelho
Genre: Fiction / Inspirational
Price: Rs 399
Description: A shepherd boy named Santiago



Atomic Habits
Author: James Clear
Genre: Self-help / Psychology
Price: Rs 450
Description: A practical guide to building good

Figure 4: Seller catalog sheet showing listed items with title, author, genre, price, and cover image.

5. Seller Orders Management

BookStore (Seller)

HomeMy ProductsAdd BooksOrdersLogout (Pravanshu)

Orders

Atomic Habits

James Clear

Product Name

Order ID

Customer Name

Address

Booking Date

Delivery By

Warranty

Price

Status

68959c59d9

Ram

173, Mayur Vihar Phase-1, Delhi (110091), Delhi

2025-08-08

2025-08-13

1 year

₹549

ontheway

The Art of War

Sun Tzu

Product Name

Order ID

Customer Name

Address

Booking Date

Delivery By

Warranty

Price

Status

68959deb8e

Akash

Balaji Medical, Datia (475661), Madhya Pradesh

2025-08-08

2025-08-13

1 year

₹498

ontheway

Figure 5: Fulfilling orders interface for sellers to trace ordered items, prices, booking dates, and update statuses.

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